# Azure Bicep in the Real World:

## Patterns, Pitfalls, and Pragmatism

DDD NORTH

# Andrew Wilson

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## Black Marble

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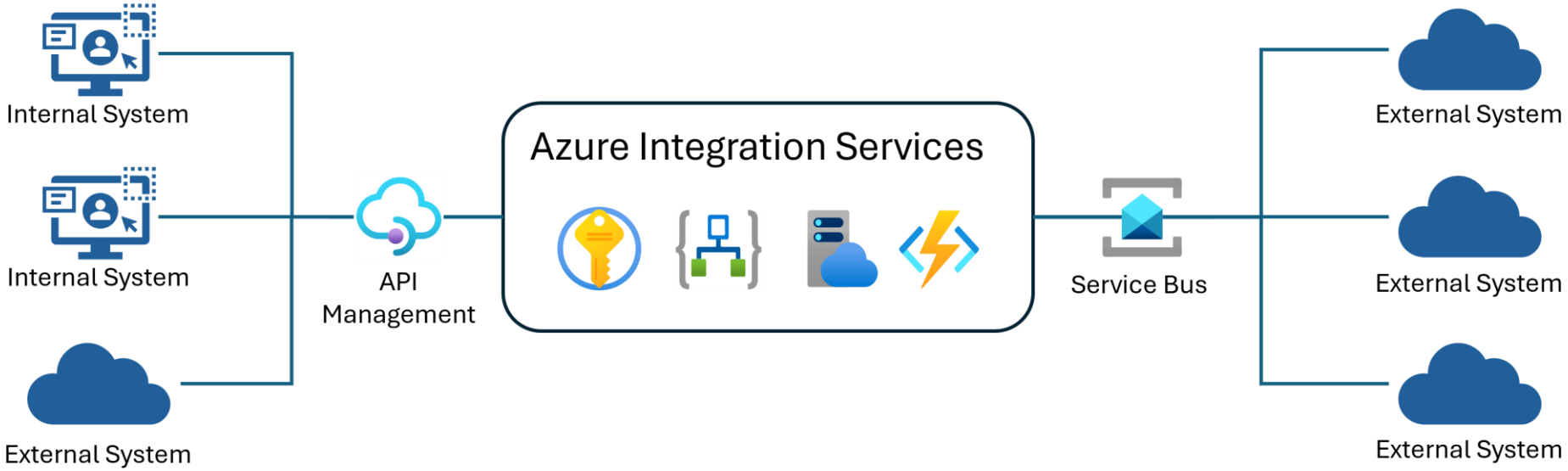
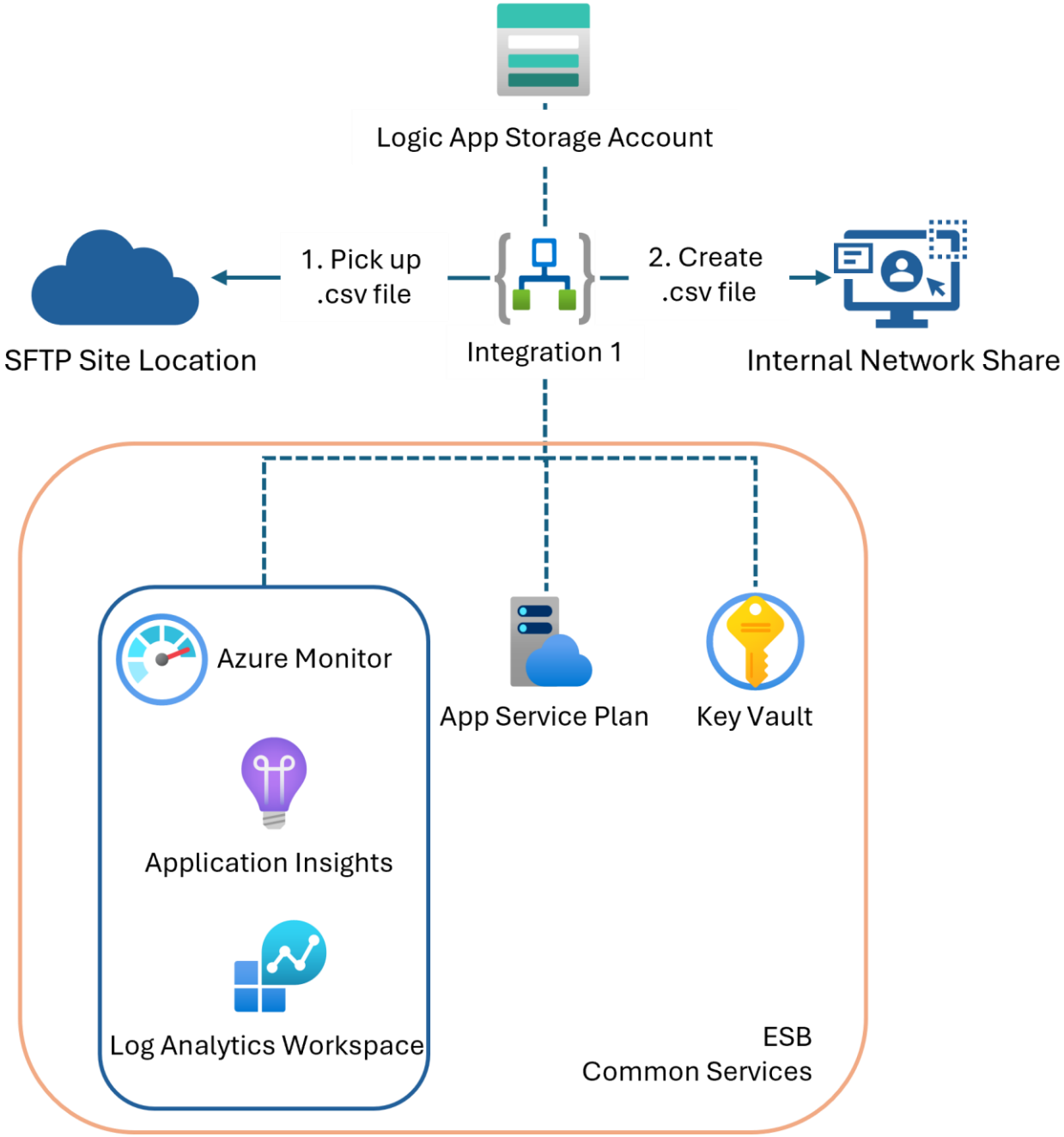
“You Are Not Alone ....”



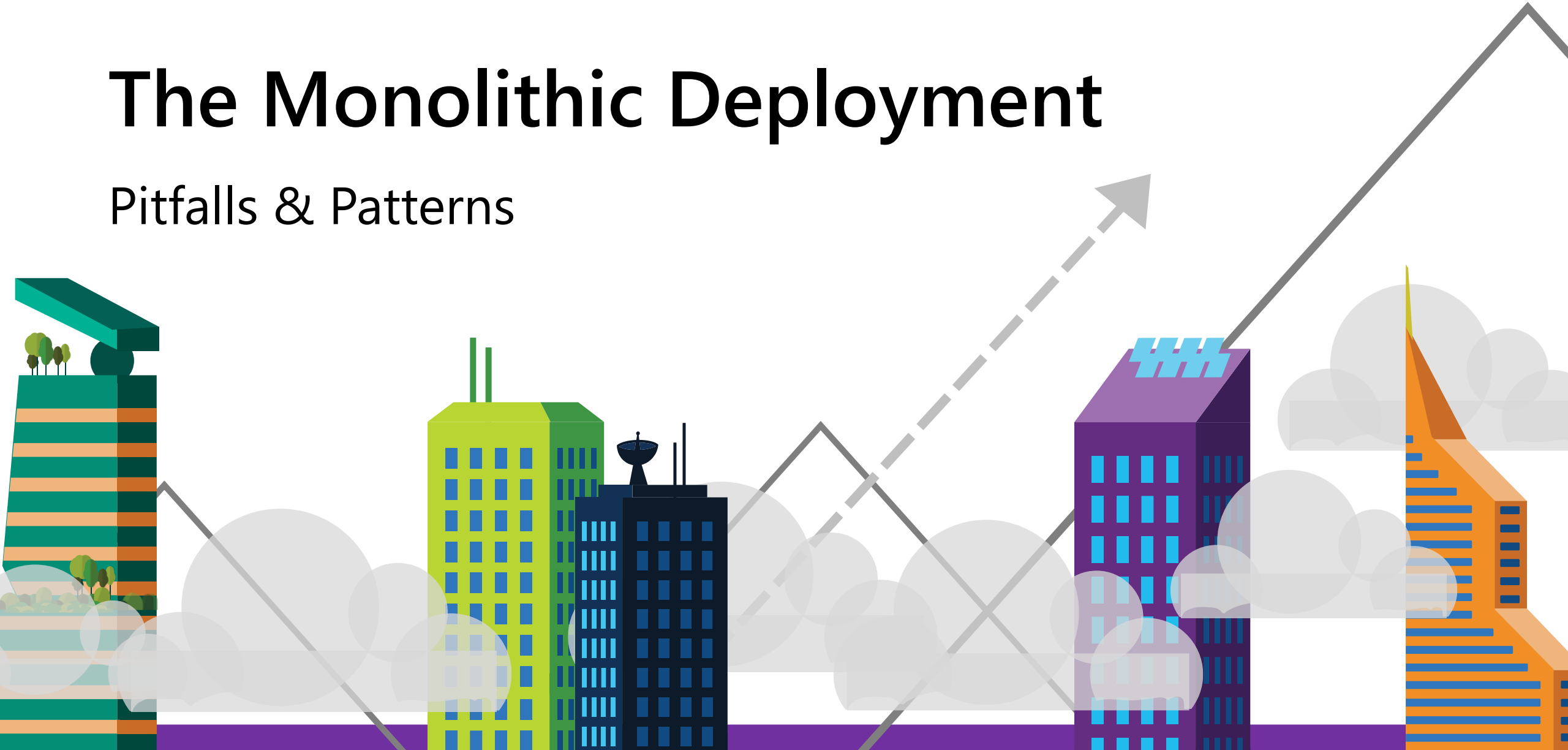

# HL Enterprise Service Bus Architecture

## The Example


Logic App Storage Account
1. Pick up .csv file
2. Create .csv file
SFTP Site Location
Integration 1
Internal Network Share
Azure Monitor
App Service Plan
Key Vault
Application Insights
Log Analytics Workspace
ESB
Common Services

# The Monolithic Deployment

## Pitfalls & Patterns

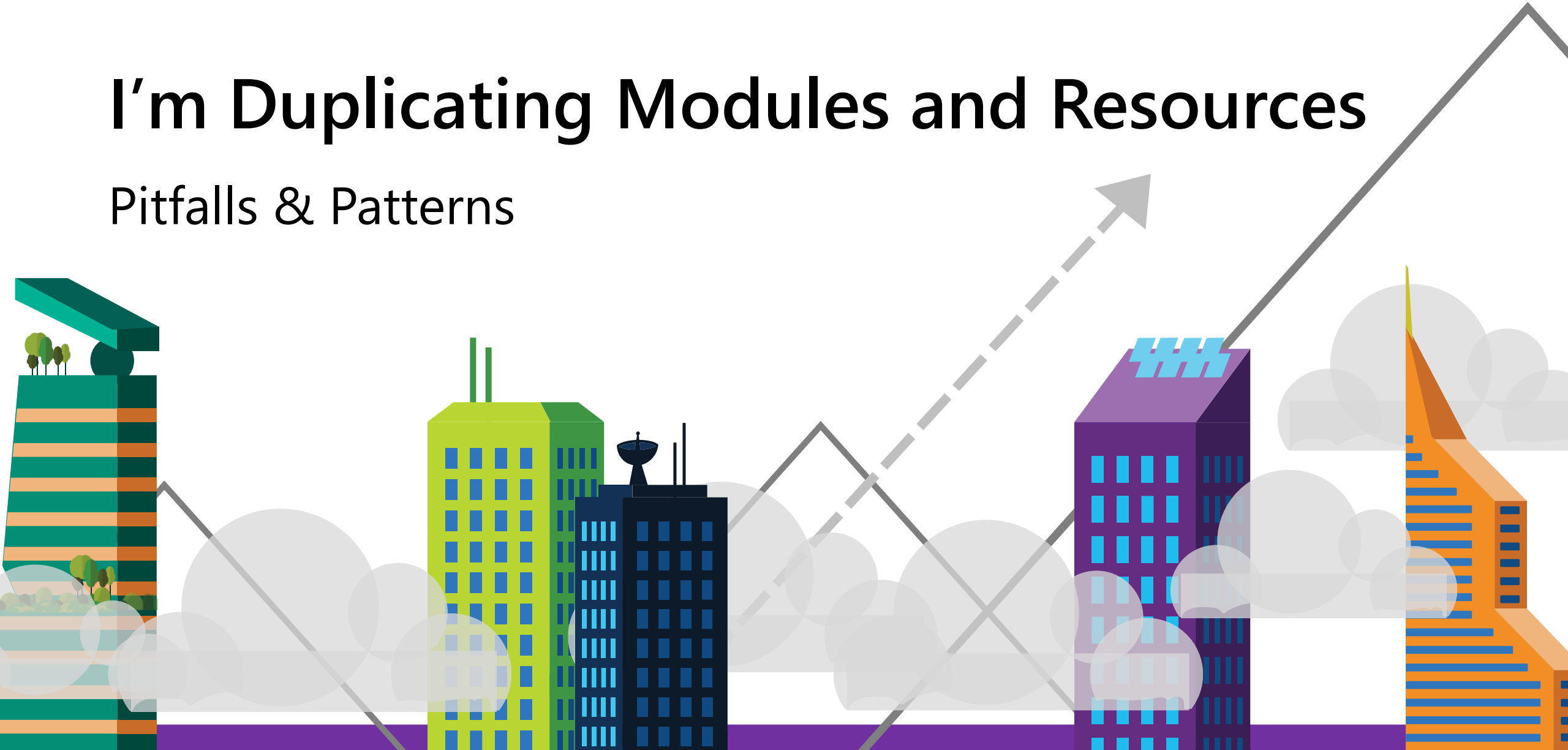


# Modular Deployments

- ESB Framework (Common – Dependency For All)
- Integration 1
- Integration 2
- Integration 3
- Integration …

# I’m Duplicating Modules and Resources

## Pitfalls & Patterns

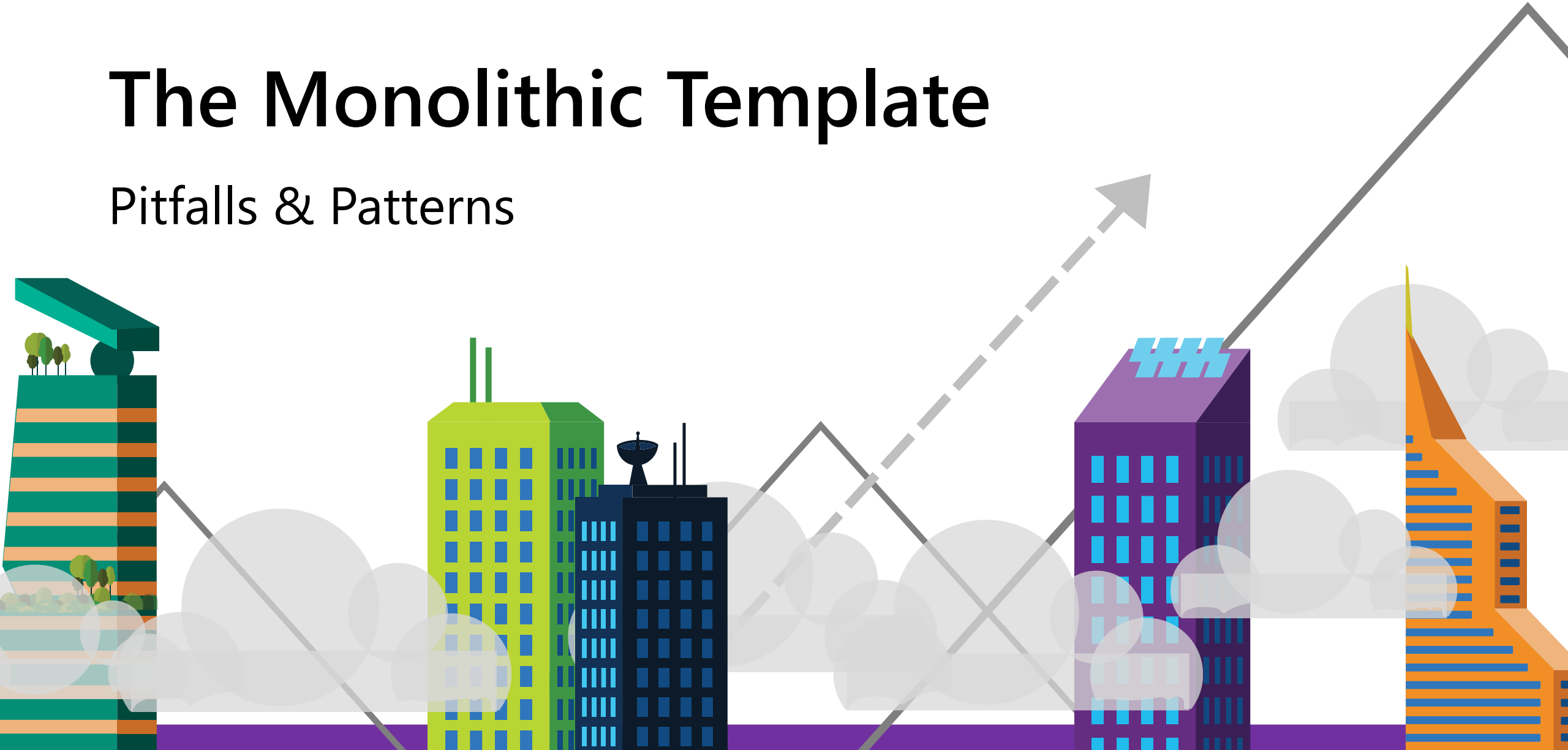


# Share Resources and Modules

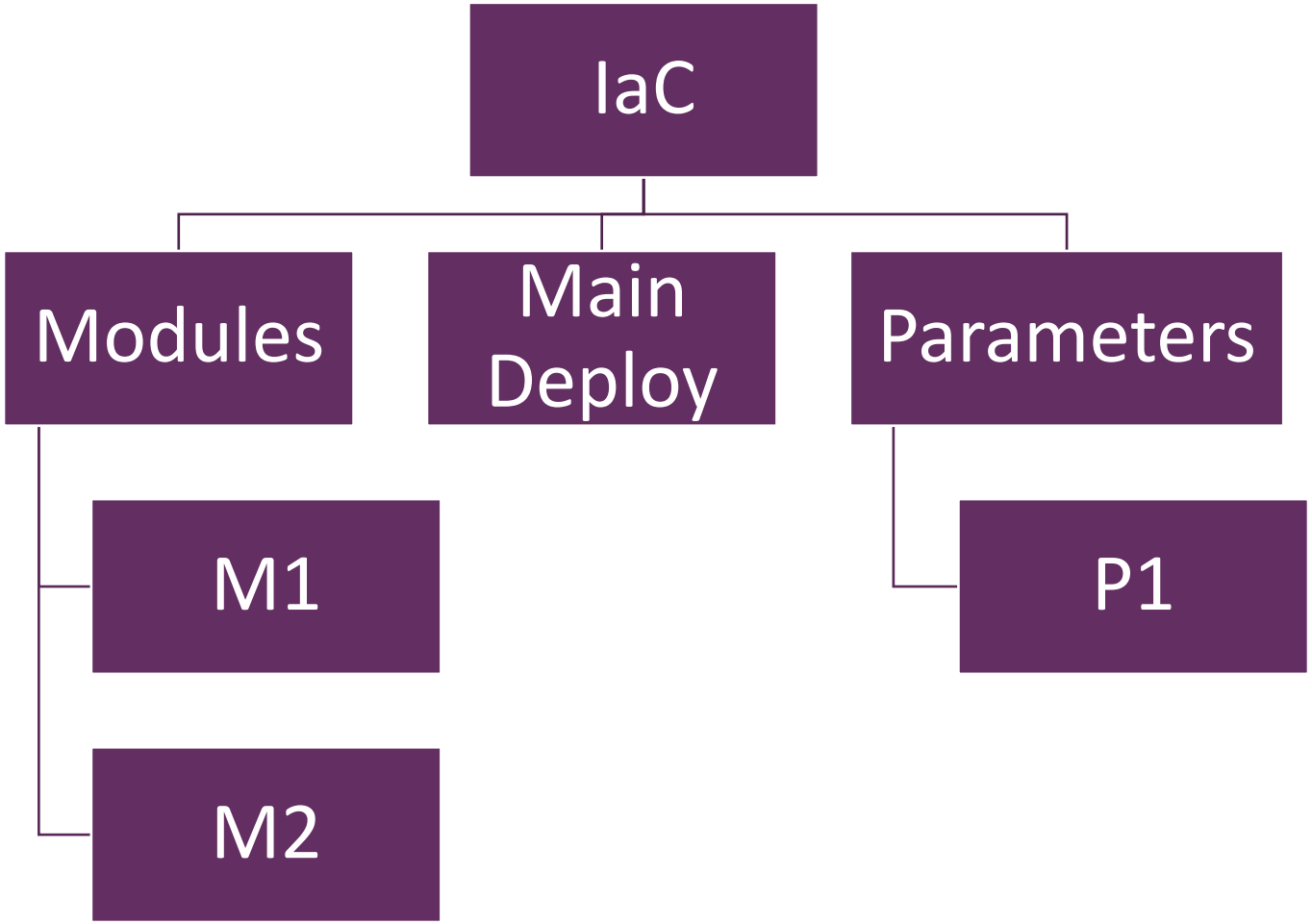
- ESB Framework (Common – Dependency For All)
- Library (Common) – Git Submodules / ACR / AVM
- Integration 1
- Integration 2
- Integration 3

# The Monolithic Template

## Pitfalls & Patterns

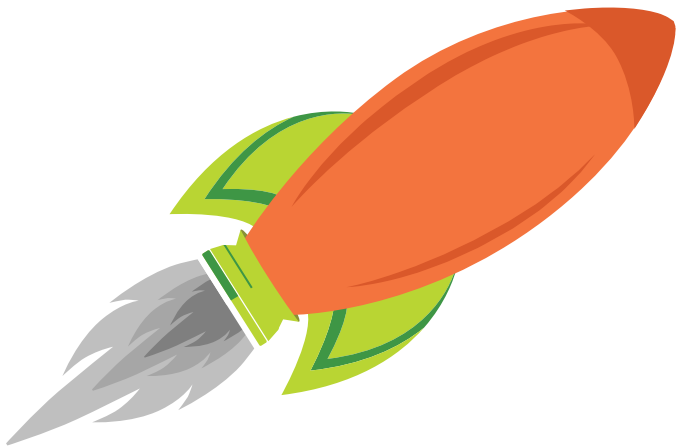


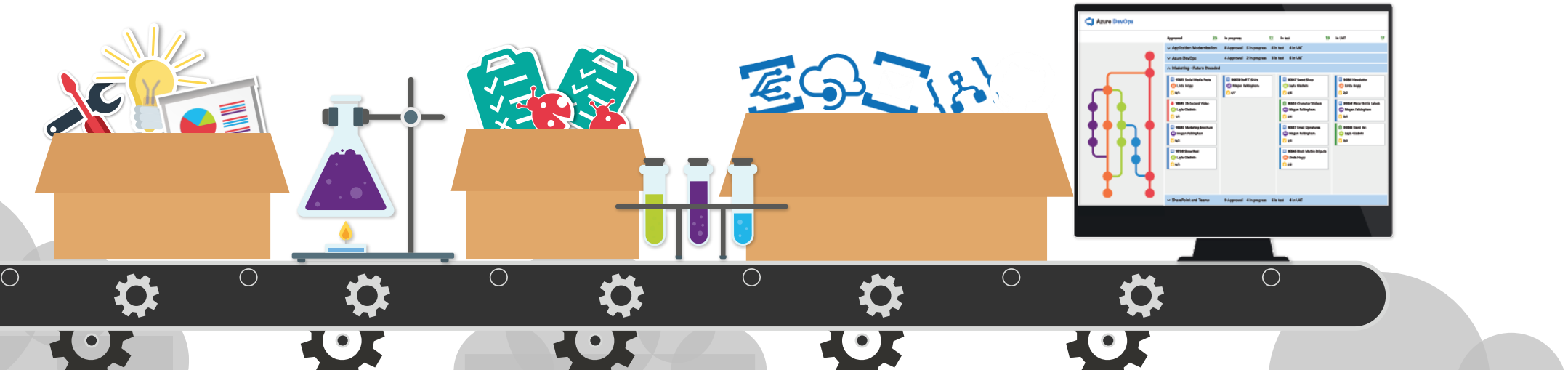
# Share Resources and Modules



# Modules

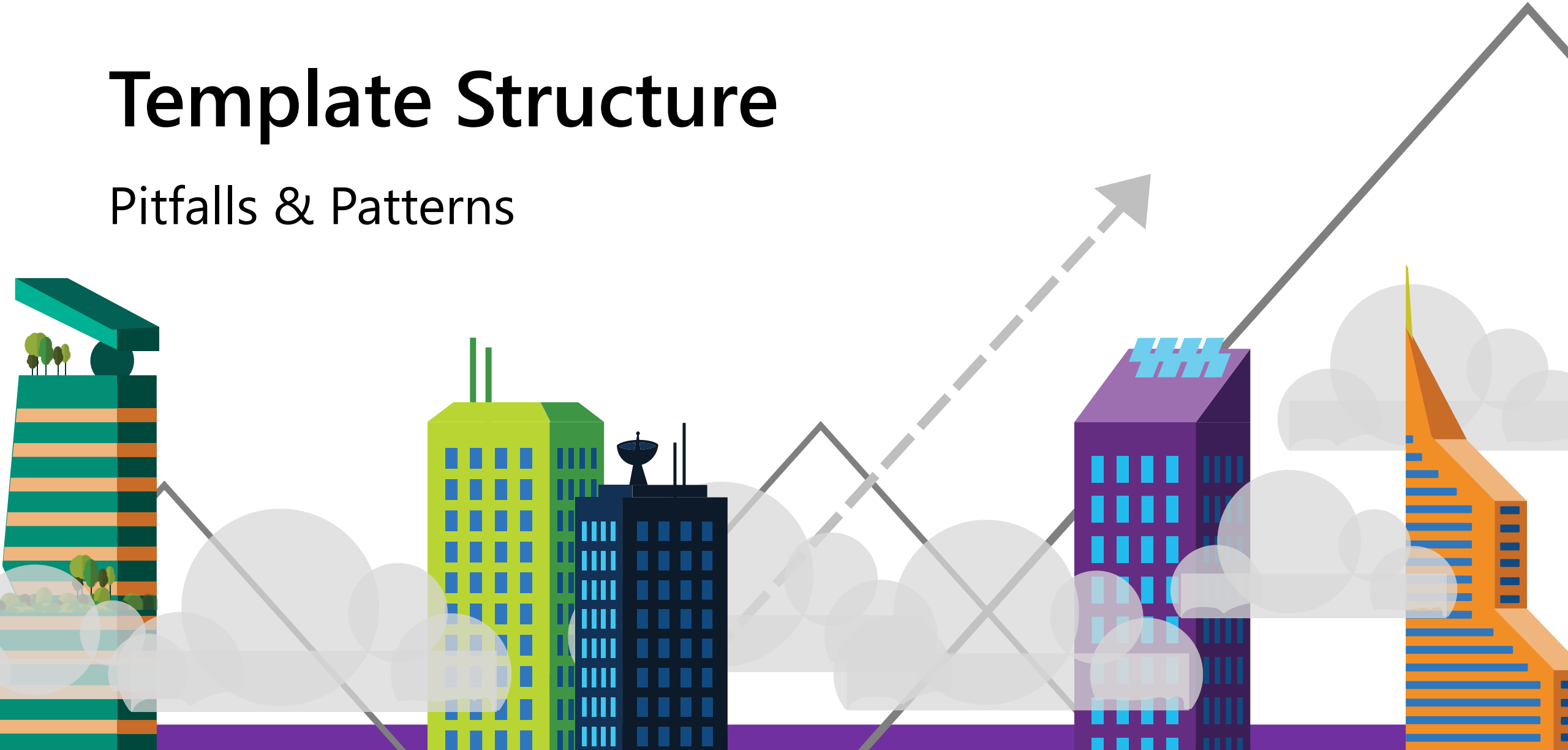
## Pitfalls & Patterns





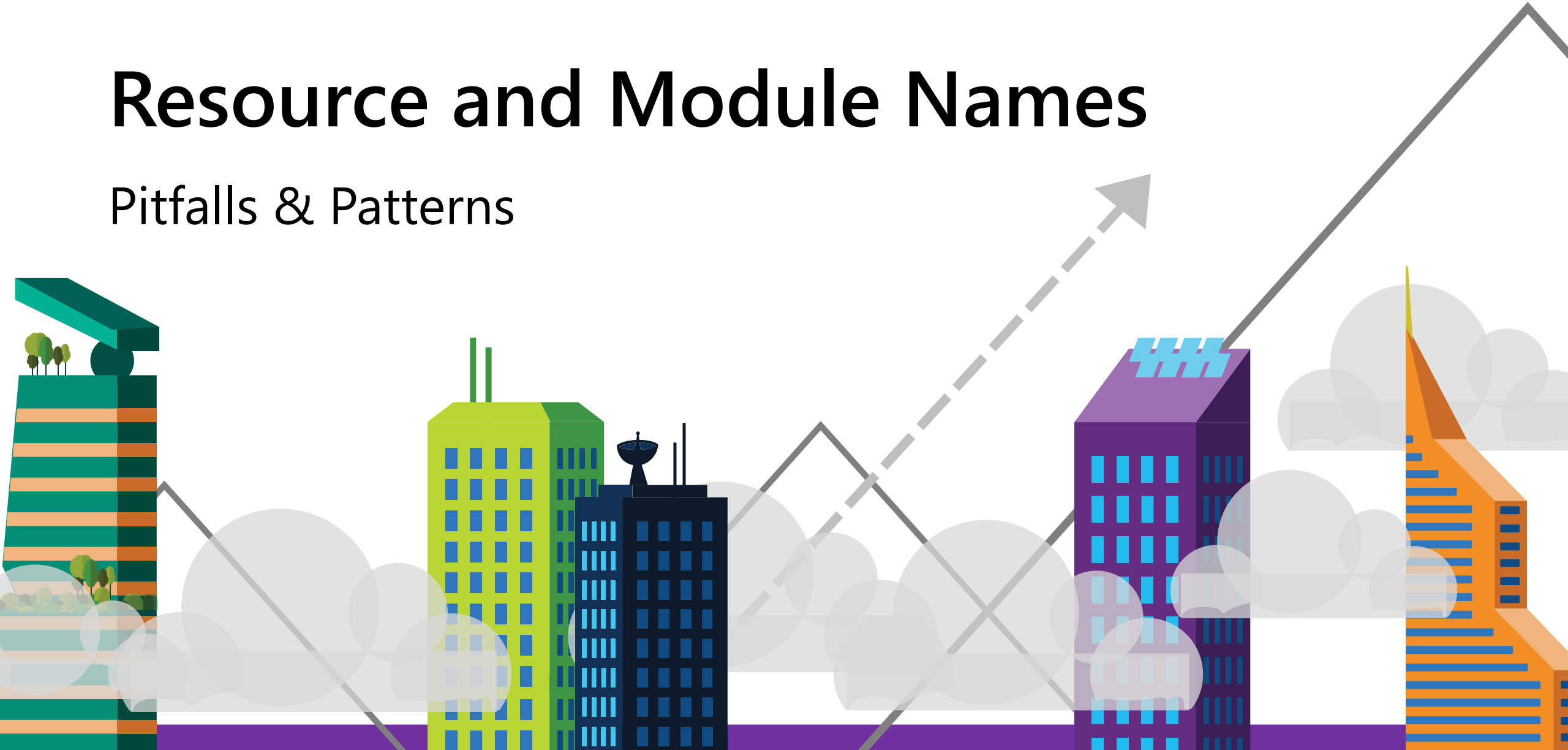
# Template Structure

Pitfalls & Patterns



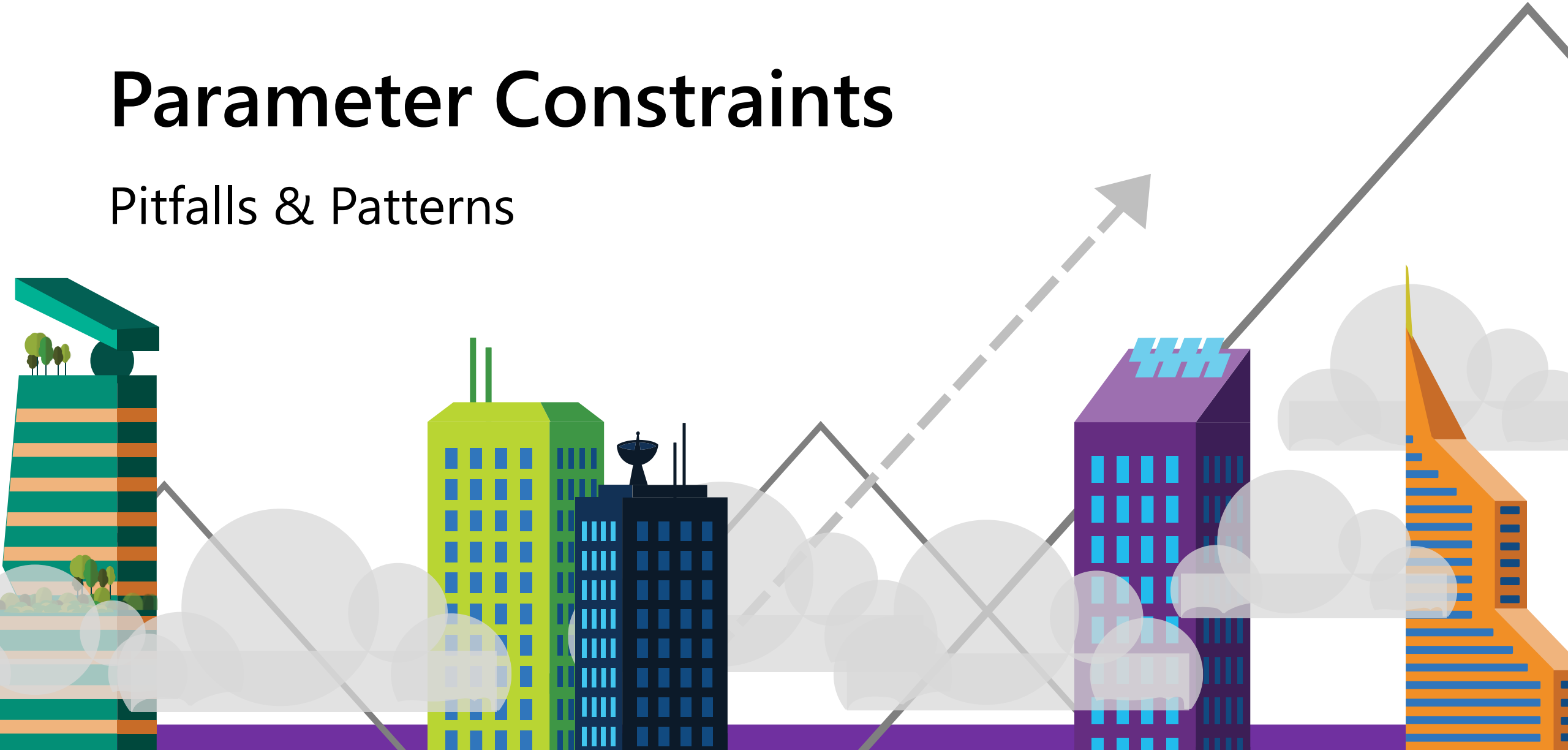
# Resource and Module Names

## Pitfalls & Patterns



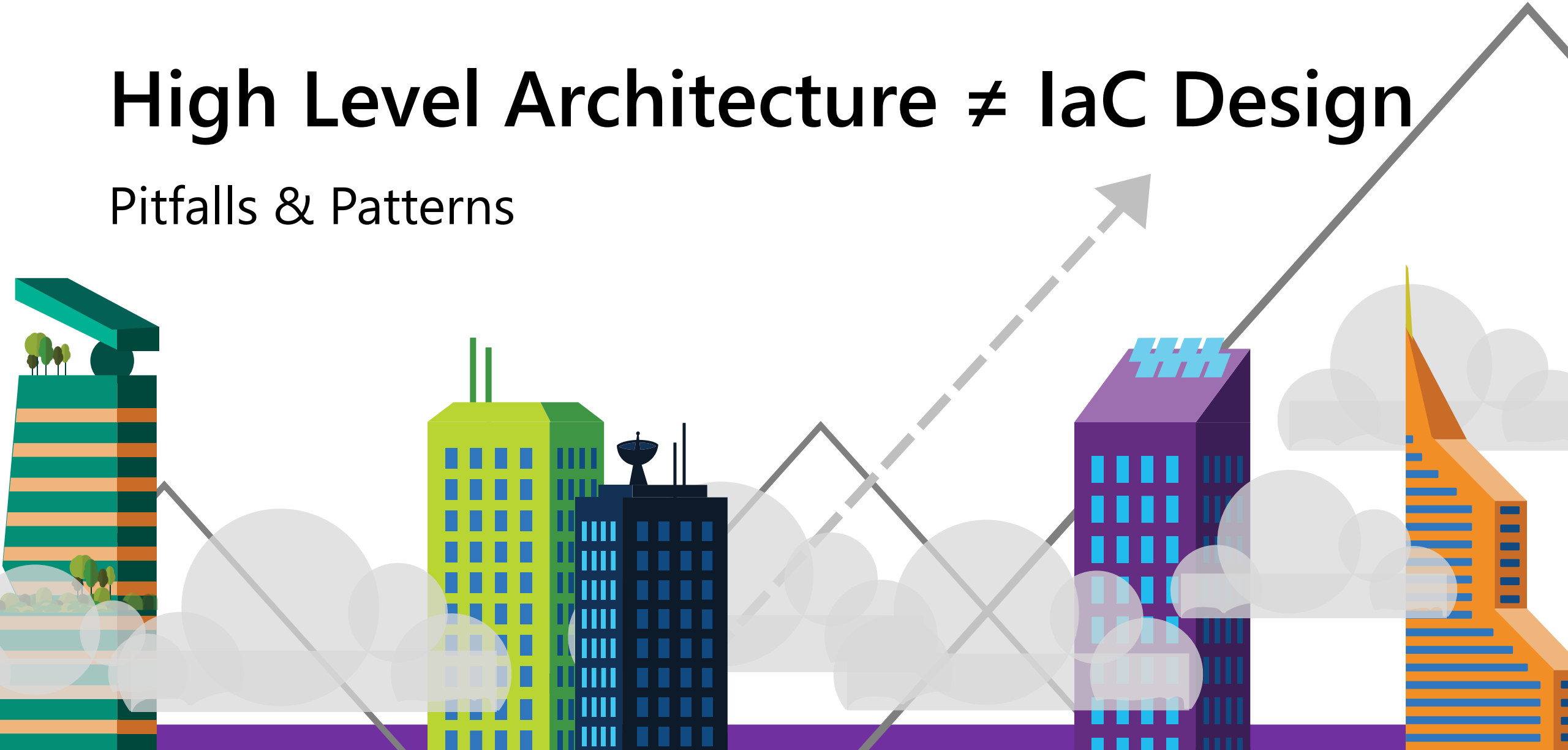
# Parameter Constraints

Pitfalls & Patterns



# High Level Architecture ≠ IaC Design

## Pitfalls & Patterns



# Design For Multiple Environments

Pitfalls & Patterns

```
resource integrationOneStorage 'Microsoft.Storage/storageAccounts@2023-05-01' = {
  name: storageName
  location: location
  tags:{
    'hidden-title': 'Storage Account for Dev'
  }
  sku: {
    name: 'Standard_LRS'
  }
  kind: 'StorageV2'
  properties: {
    accessTier: 'Hot'
    minimumTlsVersion: 'TLS1_2'
    supportsHttpsTrafficOnly: true
  }
}
```

```yaml
stages:
- stage: Build_Packages ···

- stage: 'Deploy'
  variables:
  - group: deploymentVariables
  dependsOn: 'Build_Packages'
  condition: succeeded()
  jobs:
    - deployment: Deploy
      strategy:
        runOnce:
          deploy:
            steps:
              - download: current ···

              - task: AzureResourceManagerTemplateDeployment@3 ···

              # Process the outputs from the ARM deployment and promote them to pipeline variables
              - task: PowerShell@2 ···
```

# Versioning

## Pitfalls & Patterns

# Manifest Versioning Build Tasks

Richard Fennell (Black Marble) | 6,509 installs | (20) | Free

Applies a version to various formats of manifest files based on the build number.

```yaml
- task: VersionJSONFile@3
  displayName: 'Version stamp ARM templates'
  inputs:
    Path: '$(Build.SourcesDirectory)\{YourPath}\IAC\ARMOutput'
    recursion: True
    VersionNumber: $(Build.BuildNumber) # an example versioning option but can also be options like gitversion
    useBuildNumberDirectly: False
    VersionRegex: \d+\.\d+\.\d+\.\d+
    versionForJSONFileFormat: '{1}.{2}.{3}.{4}'
    FilenamePattern: '\w+.json'
    Field: 'contentVersion' # Versioning is applied to the templates contentVersion field
    OutputVersion: OutputedVersion
```

# Shared Variables

Pitfalls & Patterns

# Function Namespaces

Pitfalls & Patterns

# environment() Function


```
{
  "name": "AzureCloud",
  "gallery": "https://gallery.azure.com/",
  "graph": "https://graph.windows.net/",
  "portal": "https://portal.azure.com",
  "graphAudience": "https://graph.windows.net/",
  "activeDirectoryDataLake": "https://datalake.azure.net/",
  "batch": "https://batch.core.windows.net/",
  "media": "https://rest.media.azure.net",
  "sqlManagement": "https://management.core.windows.net:8443/",
  "resourceManager": "https://management.azure.com/",
  "authentication": {
    "loginEndpoint": "https://login.microsoftonline.com/",
    "audiences": [ "https://management.core.windows.net/", "https://management.azure.com/" ],
    "tenant": "common",
    "identityProvider": "AAD"
  },
  "suffixes": {
    "acrLoginServer": ".azurecr.io",
    "azureDatalakeAnalyticsCatalogAndJob": "azuredatalakeanalytics.net",
    "azureDatalakeStoreFileSystem": "azuredatalakestore.net",
    "azureFrontDoorEndpointSuffix": "azurefd.net",
    "keyvaultDns": ".vault.azure.net",
    "sqlServerHostname": ".database.windows.net",
    "storage": "core.windows.net"
  }
}
```

# Naming Conventions

Pitfalls & Patterns

# Anti-Patterns

***Hard-coded resource names:***

```
name: 'MyLogicApp'
```

*Issues: Poor maintainability, readability, consistency, and convention adherence*

***Hard-coded naming variables:***

```
var lgName = 'MyLogicApp'
```

*Issues: Poor maintainability across templates, inconsistency, and no convention*

***Hard-coded naming convention:***

```
var lgName = 'logic-${project}-${environment}-${location}'
```

*Issues: Poor maintainability across templates, template inconsistency, and scattered convention logic*

**Example convention:** `{Resource Type}-{Project}-{Environment}-{Location}`

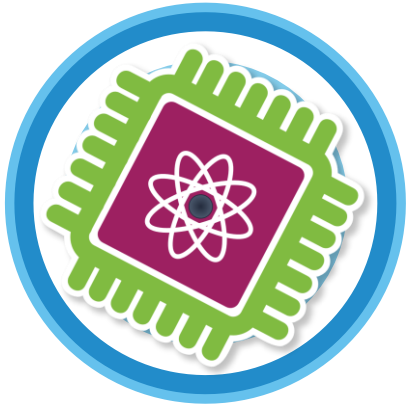
**Result:** LogicAppName: `logic-btt-dev-ukwest`

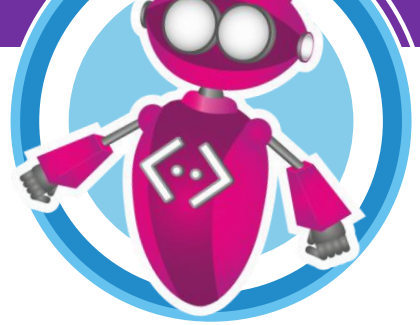
# Testing Functions and Expressions

Pitfalls & Patterns

# Secrets Must Stay Secret

Pitfalls



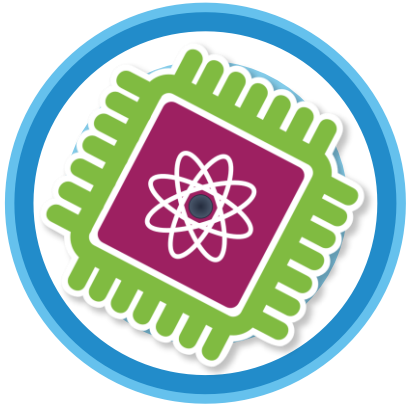


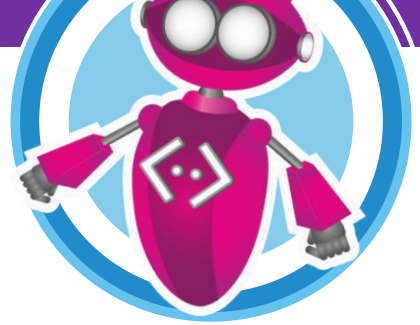
# Premature Abstraction

Anti-Patterns

# Over Specification

Anti-Patterns

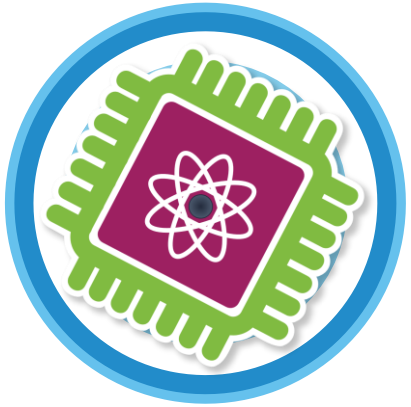


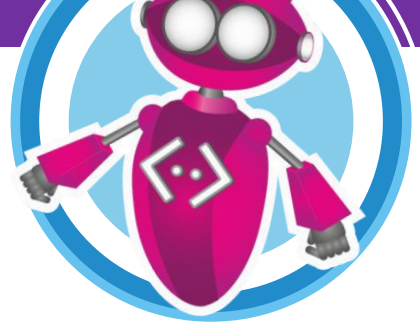


# Echo Comments

Anti-Patterns

```
// Bicep resource definition for creating a storage account
@description('Creates a storage account')
resource storageAccount 'Microsoft.Storage/storageAccounts@2021-04-01' = {
  name: 'mystorageaccount'
  location: resourceGroup().location
  sku: {
    name: 'Standard_LRS'
  }
  kind: 'StorageV2'
}
```





# Dead Code

Anti-Patterns

```
@description('Primary deployment location')
param location string = resourceGroup().location

@description('Storage account name')
param storageName string = 'st${uniqueString(resourceGroup().id)}'

@description('Never used')
param legacySku string = 'Standard_GRS'

@description('Never used')
param enableLegacyPath bool = false
```

```
// variable: Never used
var oldNamingConvention = '${storageName}-legacy'
```

```
resource storage 'Microsoft.Storage/storageAccounts@2023-05-01' = {
  name: storageName
  location: location
  sku: {
    name: 'Standard_LRS'
  }
  kind: 'StorageV2'
  tags: commonTags
  properties: {
    accessTier: 'Hot'
    minimumTlsVersion: 'TLS1_2'
    supportsHttpsTrafficOnly: true
  }
}

// resource storage2 'Microsoft.Storage/storageAccounts@2023-05-01' = {
//   name: storageName
//   location: location
//   sku: {
//     name: 'Standard_LRS'
//   }
//   kind: 'StorageV2'
//   tags: commonTags
//   properties: {
//     accessTier: 'Hot'
//     minimumTlsVersion: 'TLS1_2'
//     supportsHttpsTrafficOnly: true
//   }
// }
```

```
// variable: constant false branch
var createDeprecatedResources = false

// hardcoded false condition means this never deploys
resource deprecatedPublicIp 'Microsoft.Network/publicIPAddresses@2023-09-01' = if (createDeprecatedResources) {
  name: 'pip-${uniqueString(resourceGroup().id)}'
  location: location
  sku: {
    name: 'Basic'
  }
  properties: {
    publicIPAllocationMethod: 'Dynamic'
  }
}
```

# Copy-Paste Resource / Meta

Anti-Patterns

```
@description('Create storage account for integration 1')
resource integrationOneStorage 'Microsoft.Storage/storageAccounts@2023-05-01' = {
  name: storageName
  location: location
  sku: {
    name: 'Standard_LRS'
  }
  kind: 'StorageV2'
  tags: commonTags
  properties: {
    accessTier: 'Hot'
    minimumTlsVersion: 'TLS1_2'
    supportsHttpsTrafficOnly: true
  }
}

@description('Create storage account for integration 1')
resource integrationTwoStorage 'Microsoft.Storage/storageAccounts@2023-05-01' = {
  name: storageName
  location: location
  sku: {
    name: 'Standard_LRS'
  }
  kind: 'StorageV2'
  tags: commonTags
  properties: {
    accessTier: 'Hot'
    minimumTlsVersion: 'TLS1_2'
    supportsHttpsTrafficOnly: true
  }
}
```

# In Summary

# Thank You For Your Time

Any Questions